

Laser Diode – 5mW 650nm Red

PRODUCT ID: 1054



DESCRIPTION

You have some frickin' sharks at your lair, and you're thinking "it would be really great if I could attach some lasers to them" and then you see these little laser diode modules: what great timing! These encapsulated laser diodes are Class IIIa 5mW, with a 650nm red wavelength. They can be driven from 2.8V to 5.2V so they're great for your embedded electronics project. You can use these for laser harps, electronic 'trip wires', laser-vision guidance, and more! Simply connect power to the red and black wires and you will be lasing in no time.

Please note: this is not a laser pointer. It's a diode with integrated driver. You still need a 3 to 5VDC power supply. These diodes are 5mW output, and have a safety label marked on them. For more information on laser safety and ratings check out the Wikipedia page

Technical Details

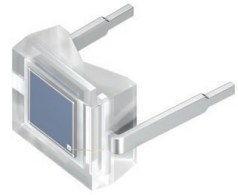


Weight: 6.3g
Diameter: 10mm / 0.4"
Length: 31mm / 1.2"
Wire length: at least 100mm / 3.9"
650nm wavelength
2.8 – 5.2V DC voltage input
25mA max current draw
Operating temperature –10°C to 40°C;

BPW 34

DIL

Silicon PIN Photodiode



Applications

- Electronic Equipment
- Industrial Automation (Machine Controls, Light Barriers, Vision Controls)

Features:

- Package: Epoxy, diffuse
- Corrosion Robustness Class: 3B
- ESD: 2 kV acc. to ANSI/ESDA/JEDEC JS-001 (HBM, Class 2)
- Especially suitable for applications from 400 nm to 1100 nm
- Short switching time (typ. 20 ns)
- DIL plastic package with high packing density

Ordering Information

Type	Photocurrent ¹⁾ $E_v = 1000 \text{ lx; Std. Light A; } V_R = 5 \text{ V}$ I_P	Photocurrent typ. $E_v = 1000 \text{ lx; Std. Light A; } V_R = 5 \text{ V}$ I_P	Ordering Code
BPW 34	$\geq 55 \mu\text{A}$	$80 \mu\text{A}$	Q62702P0073

Maximum Ratings

$T_A = 25\text{ °C}$

Parameter	Symbol		Values
Operating Temperature	T_{op}	min. max.	-40 °C 100 °C
Storage temperature	T_{stg}	min. max.	-40 °C 100 °C
Reverse voltage	V_R	max.	32 V
Total power dissipation	P_{tot}	max.	150 mW
ESD withstand voltage acc. to ANSI/ESDA/JEDEC JS-001 (HBM, Class 2)	V_{ESD}	max.	2 kV

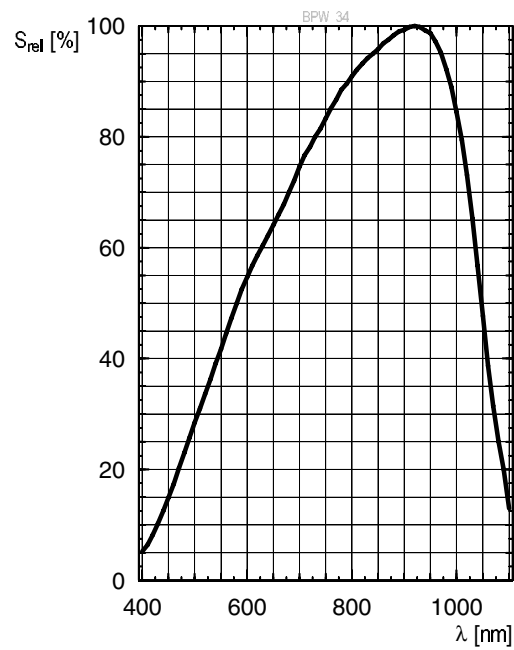
Characteristics

$T_A = 25\text{ °C}$

Parameter	Symbol		Values
Spectral sensitivity $V_R = 5\text{ V}$; Std. Light A; $T = 2856\text{ K}$	S	typ.	80 nA/lx
Wavelength of max sensitivity	$\lambda_{S\text{ max}}$	typ.	920 nm
Spectral range of sensitivity	$\lambda_{10\%}$	typ.	420 ... 1120 nm
Radiant sensitive area	A	typ.	7.02 mm ²
Dimensions of active chip area	L x W	typ.	2.65 x 2.65 mm x mm
Half angle	φ	typ.	60 °
Dark current $V_R = 10\text{ V}$	I_R	typ. max.	2 nA 30 nA
Spectral sensitivity of the chip $\lambda = 850\text{ nm}$	S_λ	typ.	0.62 A / W
Quantum yield of the chip $\lambda = 850\text{ nm}$	η	typ.	0.90 Electrons / Photon
Open-circuit voltage $E_v = 1000\text{ lx}$; Std. Light A; $V_R = 0\text{ V}$	V_O	min. typ.	300 mV 365 mV
Short-circuit current $E_v = 1000\text{ lx}$; Std. Light A; $V_R = 0\text{ V}$	I_{SC}	typ.	80 μ A
Rise time $V_R = 5\text{ V}$; $R_L = 50\text{ }\Omega$; $\lambda = 850\text{ nm}$; $I_p = 800\text{ }\mu$ A	t_r	typ.	0.02 μ s
Fall time $V_R = 5\text{ V}$; $R_L = 50\text{ }\Omega$; $\lambda = 850\text{ nm}$; $I_p = 800\text{ }\mu$ A	t_f	typ.	0.02 μ s
Forward voltage $I_F = 100\text{ mA}$; $E = 0$	V_F	typ.	1.3 V
Capacitance $V_R = 0\text{ V}$; $f = 1\text{ MHz}$; $E = 0$	C_0	typ.	72 pF
Temperature coefficient of voltage	TC_V	typ.	-2.6 mV / K
Temperature coefficient of short-circuit current Std. Light A	TC_I	typ.	0.18 % / K
Noise equivalent power $V_R = 10\text{ V}$; $\lambda = 850\text{ nm}$	NEP	typ.	0.041 pW / Hz ^{1/2}
Detection limit $V_R = 10\text{ V}$; $\lambda = 850\text{ nm}$	D^*	typ.	6.5e12 cm x Hz ^{1/2} / W

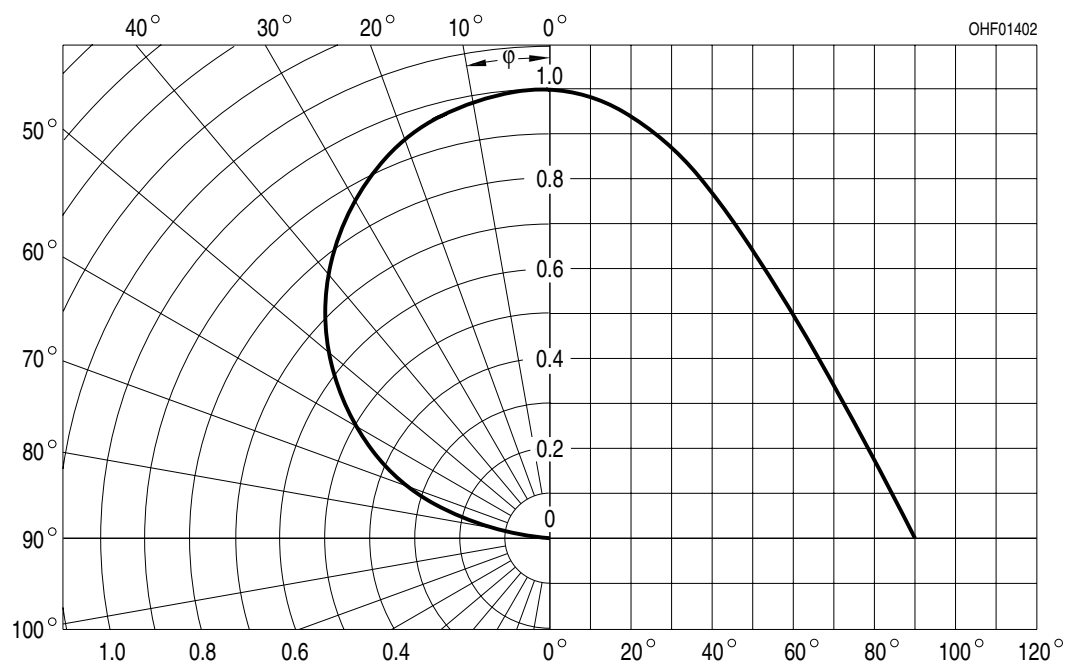
Relative Spectral Sensitivity ^{2), 3)}

$$S_{\text{rel}} = f(\lambda)$$



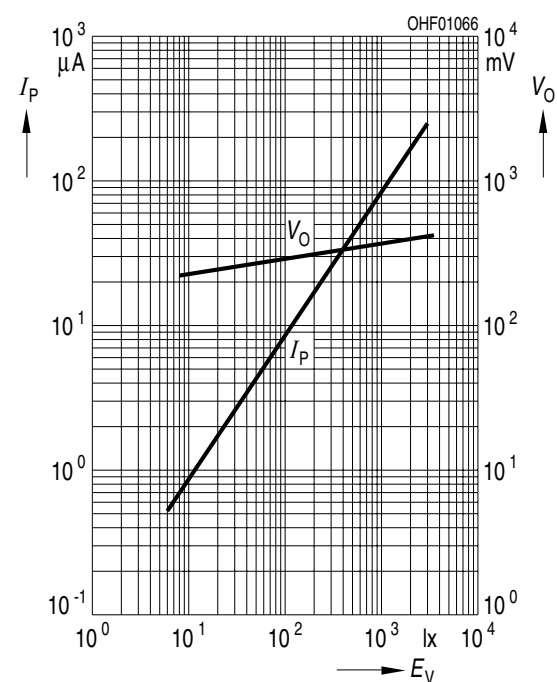
Directional Characteristics ^{2), 3)}

$$S_{\text{rel}} = f(\varphi)$$

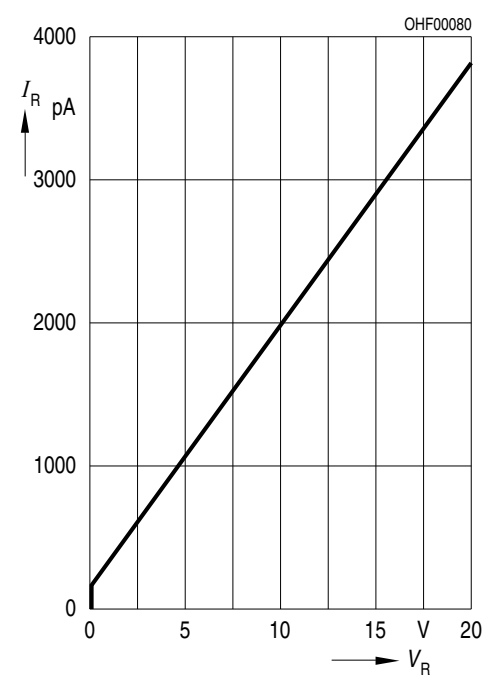


Photocurrent/Open-Circuit Voltage ^{2), 3)} Dark Current ^{2), 3)}

$$I_P (V_R = 5 \text{ V}) / V_O = f(E_e)$$

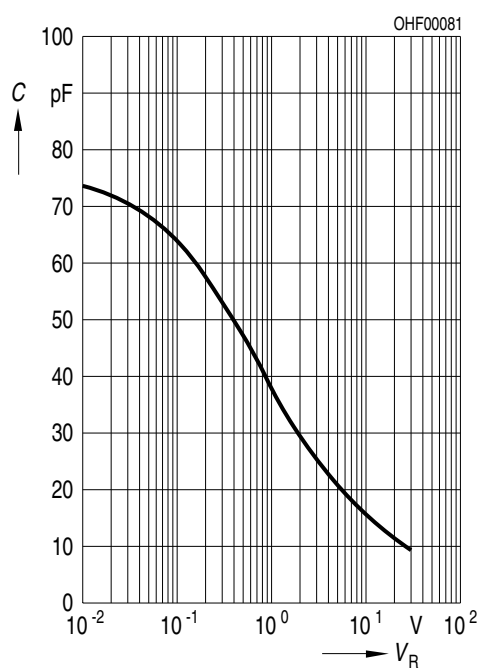


$$I_R = f(V_R); E = 0$$



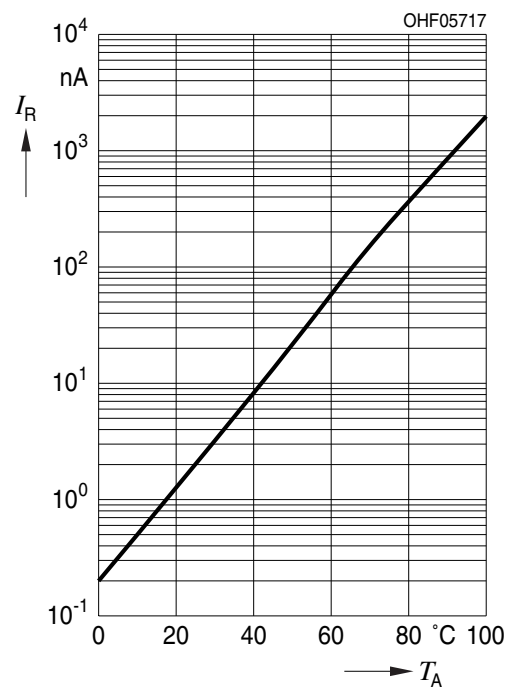
Capacitance ^{2), 3)}

$$C = f(V_R); f = 1\text{MHz}; E = 0; T_A = 25^\circ\text{C}$$



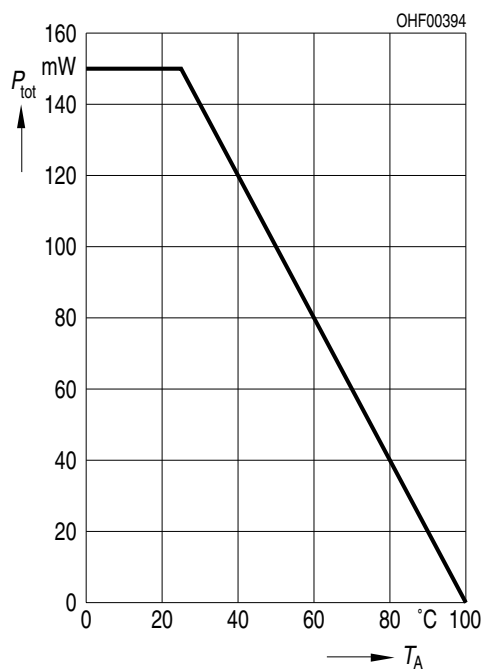
Dark Current ²⁾

$$I_R = f(T_A); E = 0; V_R = 10 \text{ V}$$

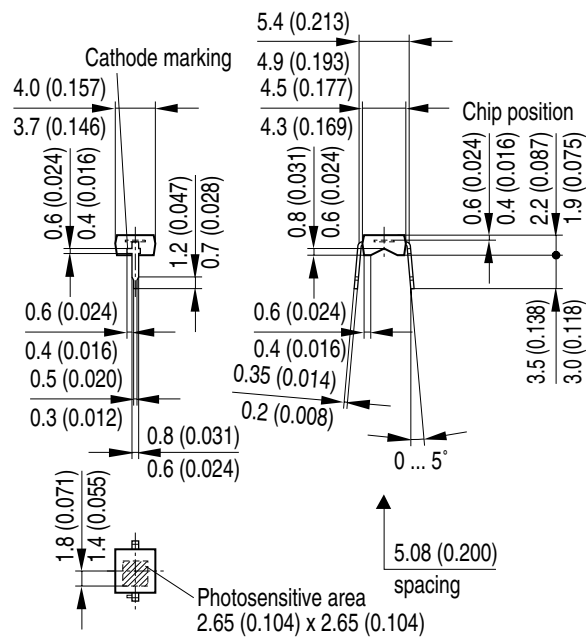


Power Consumption

$$P_{\text{tot}} = f(T_A)$$



Dimensional Drawing ⁴⁾



GEOY6643

Further Information:

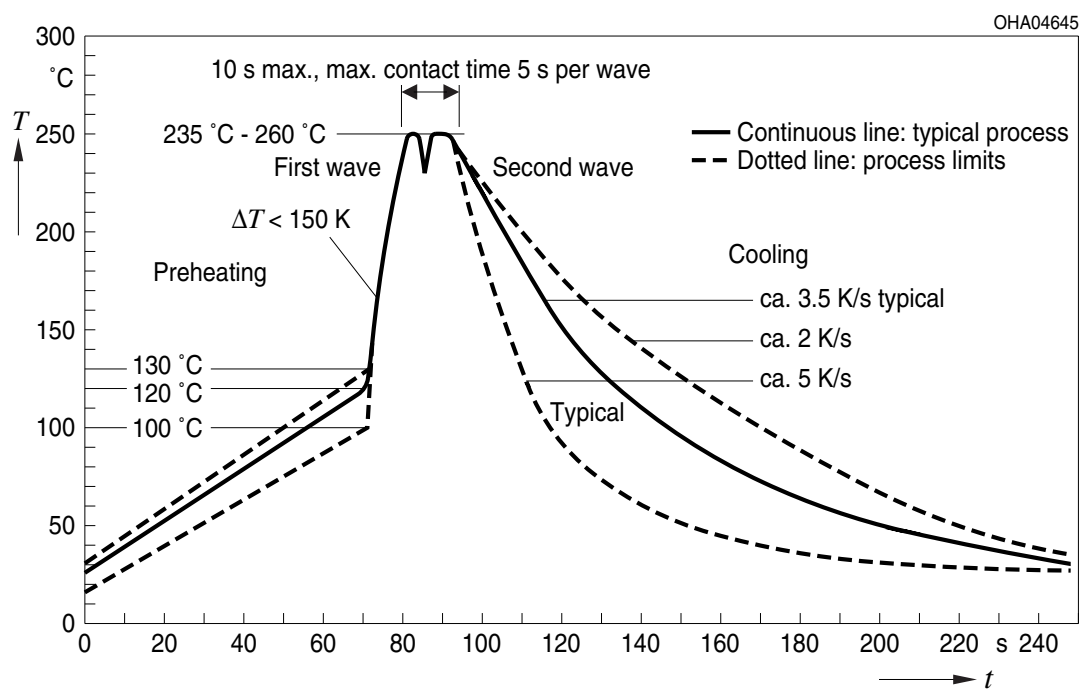
Approximate Weight: 78.0 mg

Package marking: Cathode

Corrosion test: Class: 3B
Test condition: 40°C / 90 % RH / 15 ppm H₂S / 14 days (stricter than IEC 60068-2-43)

TTW Soldering

IEC-61760-1 TTW



Notes

The evaluation of eye safety occurs according to the standard IEC 62471:2006 (photo biological safety of lamps and lamp systems). Within the risk grouping system of this IEC standard, the device specified in this data sheet falls into the class **exempt group (exposure time 10000 s)**. Under real circumstances (for exposure time, conditions of the eye pupils, observation distance), it is assumed that no endangerment to the eye exists from these devices. As a matter of principle, however, it should be mentioned that intense light sources have a high secondary exposure potential due to their blinding effect. When looking at bright light sources (e.g. headlights), temporary reduction in visual acuity and afterimages can occur, leading to irritation, annoyance, visual impairment, and even accidents, depending on the situation.

Subcomponents of this device contain, in addition to other substances, metal filled materials including silver. Metal filled materials can be affected by environments that contain traces of aggressive substances. Therefore, we recommend that customers minimize device exposure to aggressive substances during storage, production, and use. Devices that showed visible discoloration when tested using the described tests above did show no performance deviations within failure limits during the stated test duration. Respective failure limits are described in the IEC60810.

For further application related information please visit www.osram-os.com/appnotes

Disclaimer

Attention please!

The information describes the type of component and shall not be considered as assured characteristics. Terms of delivery and rights to change design reserved. Due to technical requirements components may contain dangerous substances.

For information on the types in question please contact our Sales Organization.

If printed or downloaded, please find the latest version on the OSRAM OS website.

Packing

Please use the recycling operators known to you. We can also help you – get in touch with your nearest sales office. By agreement we will take packing material back, if it is sorted. You must bear the costs of transport. For packing material that is returned to us unsorted or which we are not obliged to accept, we shall have to invoice you for any costs incurred.

Product and functional safety devices/applications or medical devices/applications

OSRAM OS components are not developed, constructed or tested for the application as safety relevant component or for the application in medical devices.

OSRAM OS products are not qualified at module and system level for such application.

In case buyer – or customer supplied by buyer – considers using OSRAM OS components in product safety devices/applications or medical devices/applications, buyer and/or customer has to inform the local sales partner of OSRAM OS immediately and OSRAM OS and buyer and /or customer will analyze and coordinate the customer-specific request between OSRAM OS and buyer and/or customer.

Glossary

- ¹⁾ **Photocurrent:** The photocurrent values are measured (by irradiating the devices with a homogenous light source and applying a voltage to the device) with a tolerance of $\pm 11\%$.
- ²⁾ **Typical Values:** Due to the special conditions of the manufacturing processes of semiconductor devices, the typical data or calculated correlations of technical parameters can only reflect statistical figures. These do not necessarily correspond to the actual parameters of each single product, which could differ from the typical data and calculated correlations or the typical characteristic line. If requested, e.g. because of technical improvements, these typ. data will be changed without any further notice.
- ³⁾ **Testing temperature:** $T_A = 25^\circ\text{C}$ (unless otherwise specified)
- ⁴⁾ **Tolerance of Measure:** Unless otherwise noted in drawing, tolerances are specified with ± 0.1 and dimensions are specified in mm.

Revision History

Version	Date	Change
1.5	2020-01-07	Characteristics Electro - Optical Characteristics (Diagrams)

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ARDUINO UNO REV3

Code: A000066

The UNO is the best board to get started with electronics and coding. If this is your first experience tinkering with the platform, the UNO is the most robust board you can start playing with. The UNO is the most used and documented board of the whole Arduino family.

- **Arduino Uno** is a microcontroller board based on the ATmega328P (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz quartz crystal, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.. You can tinker with your UNO without worrying too much about doing something wrong, worst case scenario you can replace the chip for a few dollars and start over again.

"Uno" means one in Italian and was chosen to mark the release of Arduino Software (IDE) 1.0. The Uno board and version 1.0 of Arduino Software (IDE) were the reference versions of Arduino, now evolved to newer releases. The Uno board is the first in a series of USB Arduino boards, and the reference model for the Arduino platform; for an extensive list of current, past or outdated boards see the Arduino index of boards.

You can find here your board warranty informations.
<https://www.arduino.cc/en/Main/warranty>

Getting Started

You can find in the Getting Started section all the information you need to configure your board, use the Arduino Software (IDE), and start tinker with coding and electronics.
<https://www.arduino.cc/en/Guide/HomePage>

TECH SPECS

Microcontroller	ATmega328P
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limit)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
PWM Digital I/O Pins	6
Analog Input Pins	6
DC Current per I/O Pin	20 mA
DC Current for 3.3V Pin	50 mA

Flash Memory	32 KB (ATmega328P) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328P)
EEPROM	1 KB (ATmega328P)
Clock Speed	16 MHz
LED_BUILTIN	13
Length	68.6 mm
Width	53.4 mm
Weight	25 g

• OSH: Schematics

Arduino Uno is open-source hardware! You can build your own board using the following files:

EAGLE FILES IN .ZIP

https://www.arduino.cc/en/uploads/Main/arduino_Uno_Rev3-02-TH.zip

SCHEMATICS IN .PDF

https://www.arduino.cc/en/uploads/Main/Arduino_Uno_Rev3-schematic.pdf

BOARD SIZE IN .DXF

<http://arduino.cc/documents/ArduinoUno.dxf>

Programming

The Arduino Uno can be programmed with the (Arduino Software (IDE)). Select "Arduino/Genuino Uno from the Tools > Board menu (according to the microcontroller on your board). For details, see the reference and tutorials.

The ATmega328 on the Arduino Uno comes preprogrammed with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files).

You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header using Arduino ISP or similar; see these instructions for details.

The ATmega16U2 (or 8U2 in the rev1 and rev2 boards) firmware source code is available in the Arduino repository. The ATmega16U2/8U2 is loaded with a DFU bootloader, which can be activated by:

- On Rev1 boards: connecting the solder jumper on the back of the board (near the map of Italy) and then resetting the 8U2.
- On Rev2 or later boards: there is a resistor that pulling the 8U2/16U2 HWB line to ground, making it easier to put into DFU mode.

You can then use Atmel's FLIP software (Windows) or the DFU programmer (Mac OS X and Linux) to load a new firmware. Or you can use the ISP header with an external programmer (overwriting the DFU bootloader). See this user-contributed tutorial for more information.

Warnings

The Arduino Uno has a resettable polyfuse that protects your computer's USB ports from shorts and overcurrent. Although most computers provide their own internal protection, the fuse provides an extra layer of protection. If more than 500 mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed.

Differences with other boards

The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the ATmega16U2 (ATmega8U2 up to version R2) programmed as a USB-to-serial converter.

Power

The Arduino Uno board can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the GND and Vin pin headers of the POWER connector.

The board can operate on an external supply from 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may become unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The power pins are as follows:

- Vin. The input voltage to the Arduino/Genuino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- 5V. This pin outputs a regulated 5V from the regulator on the board. The board can be supplied with power either from the DC power jack (7 - 12V), the USB connector (5V), or the VIN pin of the board (7-12V). Supplying voltage via the 5V or 3.3V pins bypasses the regulator, and can damage your board. We don't advise it.
- 3V3. A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- GND. Ground pins.
- IOREF. This pin on the Arduino/Genuino board provides the voltage reference with which the microcontroller operates. A properly configured shield can read the IOREF pin voltage and select the appropriate power source or enable voltage translators on the outputs to work with the 5V or 3.3V.

Memory

The ATmega328 has 32 KB (with 0.5 KB occupied by the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

<https://www.arduino.cc/en/Reference/EEPROM>

Input and Output

See the mapping between Arduino pins and ATmega328P ports. The mapping for the Atmega8, 168, and 328 is identical.

PIN MAPPING ATmega328P

<https://www.arduino.cc/en/Hacking/PinMapping168>

Each of the 14 digital pins on the Uno can be used as an input or output, using `pinMode()` <https://www.arduino.cc/en/Reference/PinMode>, `digitalWrite()` <https://www.arduino.cc/en/Reference/DigitalWrite>, and `digitalRead()` <https://www.arduino.cc/en/Reference/DigitalRead> functions. They operate at 5 volts. Each pin can provide or receive 20 mA as recommended operating condition and has an internal pull-up resistor (disconnected by default) of 20-50k ohm. A maximum of 40mA is the value that must not be exceeded on any I/O pin to avoid permanent damage to the microcontroller.

In addition, some pins have specialized functions:

- Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
- External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the `attachInterrupt()` function for details.
- PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the `analogWrite()` function.
- SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK). These pins support SPI communication using the SPI library.
- LED: 13. There is a built-in LED driven by digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.
- TWI: A4 or SDA pin and A5 or SCL pin. Support TWI communication using the Wire library.

The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though it is possible to change the upper end of their range using the AREF pin and the `analogReference()` function. There are a couple of other pins on the board:

- AREF. Reference voltage for the analog inputs. Used with `analogReference()`.
- Reset. Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

Communication

Arduino/Genuino Uno has a number of facilities for communicating with a computer, another Arduino/Genuino board, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The 16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino Software (IDE) includes a serial monitor which allows simple textual data to be sent to and from the board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1).

A SoftwareSerial library allows serial communication on any of the Uno's digital pins.

<https://www.arduino.cc/en/Reference/SoftwareSerial>

The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino Software (IDE) includes a Wire library to simplify use of the I2C bus; see the documentation for details. For SPI communication, use the SPI library.

<https://www.arduino.cc/en/Reference/Wire>

<https://www.arduino.cc/en/Reference/SPI>

Automatic (Software) Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino/Genuino Uno board is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino Software (IDE) uses this capability to allow you to upload code by simply pressing the upload button in the interface toolbar. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.

This setup has other implications. When the Uno is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half-second or so, the bootloader is running on the Uno. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts, make sure that the software with which it communicates waits a second after opening the connection and before sending this data.

The Uno board contains a trace that can be cut to disable the auto-reset. The pads on either side of the trace can be soldered together to re-enable it. It's labeled "RESET-EN". You may also be able to disable the auto-reset by connecting a 110 ohm resistor from 5V to the reset line; see this forum thread for details.

Revisions

Revision 3 of the board has the following new features:

- 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage

provided from the board. In future, shields will be compatible with both the board that uses the AVR, which operates with 5V and with the Arduino Due that operates with 3.3V. The second one is a not connected pin, that is reserved for future purposes.

- Stronger RESET circuit.
- Atmega 16U2 replace the 8U2.





Fibre Optic Cleaning Tools & Test Equipment

Visual Fault Locator (VFL) 30Mw - Model 01

Product Description

The Briticom® Visual Fault Locator (VFL) is an essential tool for any engineer or technician. This allows quick and easy detection of faults in fibre installations, such as breaks or faulty connectors, by pinpointing the location of the signal loss. Therefore, problems can be diagnosed and fixed efficiently. The VFL can also be used for end to end continuity tests and perform fibre identification.

Features & Benefits

- Quick & easy location of damage in fibres
- Continuous & flashing illumination models
- Designed for demanding field use
- Long-lasting battery life
- Protective rubber end cap
- Soft-sided carry case with belt loop
- One-year warranty

Applications

- Fibre optic inspection and testing



Technical Specifications

Wavelength	650nm
Emitter Type	FP-LD source
Output Power	30mW max
Distance Range	30km (single-mode); >5km (multi-mode)
Operation Mode	Pulsed & CW
Power Supply	2x AA batteries
Material shell	Alum-alloy
Connector Type	Universal 2.5mm
Dimensions (mm)	175x26(Ø)
Weight (g)	173

To Order

To place an order, use the contact details below.





Fibre Optic Cleaning Tools & Test Equipment

Visual Fault Locator (VFL) 30Mw - Model 01

Ordering Information

Use combinations of **bold** letters and numbers to create part numbers for ordering.

Product		Distance	
VFL	Visual Fault Locator	7K 20K 30K	7 kilometers 15-20 kilometers 30 kilometers
▲		▲	
VFL	30	20K	M1
	▼		▼
1mW 20mW 30mW	1 20 30	Model 1 Model 2 Model 3 Model 4	M1 M2 M3 M4
Laser Power		Model	

Example Part Number	
Description	Part Number
F0 Visual Fault Locator 30mw 20km SM 5km MM	VFL3020KM1

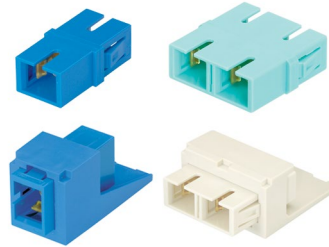
To Order

To place an order, use the contact details below.

SC Fiber Optic Adapters

specifications

SC fiber optic adapters with integrated panel retention clips are TIA/EIA-604 FOCIS-3 compliant. Each SC simplex adapter shall connect one SC connector pair in one module space. Each SC duplex adapter shall connect two SC connector pairs in two module spaces. SC adapters and adapter modules shall include phosphor bronze split sleeves for multimode applications or zirconia ceramic split sleeves for singlemode applications.



technical information

Standards requirements:	TIA/EIA-604 FOCIS-3 compliant; exceeds TIA/EIA-568-B.3 requirements
Split sleeve material:	Phosphor bronze or zirconia ceramic (required for singlemode applications)
Insertion loss:	0.1dB average (multimode), 0.15dB average (singlemode); supports the performance of FOCIS-3 compliant connectors/patch cords
Return loss:	Supports singlemode (>40dB for SPC and >55dB for UPC) and multimode (>20dB) connector polish performance

*Panduit LC Connector/Adapter connection

key features and benefits

Integrated panel retention clips:	No metal clips to become bent, damaged or lost; clips automatically adjust for FOCIS-3 compliant panel thickness variations for improved mounting
High glass transition temperature latch material (Tg>100°C)	Meets Telcordia GR-326-CORE, Issue 3; maintains latch integrity/geometry in high temperature environments for improved reliability
Improved adapter protective cap design	Maximum protection from contamination; protective cap fully surrounds split sleeve opening
Short flange design	Improved modularity and higher density usage
Choice of phosphor bronze or zirconia ceramic split sleeves	Provides an adapter split sleeve material option to fit specific network requirements
Q.C. number and split sleeve material laser marked on every adapter	Assures 100% traceability; quick and easy identification of split sleeve material: SC-P = Phosphor Bronze and SC-Z = Zirconia Ceramic
Adapters available in Mini-Com® Modules	System flexibility; adapter modules can be used in all Mini-Com® Closet Products and Work Area Outlets
Adapters available in patch panels, fiber adapter panels (FAPs) and Opticom® QuickNet™ Pre-Terminated MTP® Cassettes	Provides a complete SC system solution
Adapters and modules available in multiple colors	Allows color-coding for network segregation

applications

Panduit SC Fiber Optic Adapters can be used with Mini-Com® Modules, Patch Panels, Faceplates and Surface Mount Boxes for a complete SC system solution. SC adapters provide a robust solution for LANs, public networks, storage area networks

*MTP is a registered trademark of US Conec Ltd.

and fiber-to-the-desk applications. SC adapters can also be mounted in high-density applications within Opticom® QuickNet™ Pre-Terminated MTP® Cassettes and Fiber Enclosures, and Opticom® Fiber Adapter Panels (FAPs) and Enclosures.

PANDUIT®

SPECIFICATION SHEET

SC Fiber Optic Adapters

Simplex MM:	FASSCEI-L
Duplex MM:	FADSCEI-L
Simplex 10Gig™:	FASSCAQ-L
Duplex 10Gig™:	FADSCAQ-L
Simplex 10Gig™ (zirc.):	FASSCZAQ-L
Duplex 10Gig™ (zirc.):	FADSCZAQ-L
Simplex SM (zirc.):	FASSCZBU-L
Duplex SM (zirc.):	FADSCZBU-L
Simplex APC (zirc.):	FASSCZAG-L
Duplex APC (zirc.):	FADSCZAG-L

L = Bag of 50 adapters; 100 per carton.

Mini-Com® SC Adapter Modules

Simplex MM:	CMSEISCEI
Duplex MM:	CMDEISCEI
Simplex 10Gig™:	CMSAQSCBL
Duplex 10Gig™:	CMDAQSCBL
Simplex 10Gig™ (zirc.):	CMSAQSCZBL
Duplex 10Gig™ (zirc.):	CMDAQSCZBL
Simplex SM (zirc.):	CMSBUSCZBU
Duplex SM (zirc.):	CMDBUSCZBU
Simplex APC (zirc.):	CMSAGSCZBL
Duplex APC (zirc.):	CMDAGSCZBL

Opticom® SC Fiber Adapter Panels

6 duplex MM:	FAP6WEIDSC
6 duplex 10Gig™:	FAP6WAQDSCZ
6 duplex SM (zirc.):	FAP6WBUDSCZ
12 simplex SM (zirc.):	FAP12WBUSCZ

Opticom® QuickNet™ Pre-Terminated MTP® Cassettes

12 SC to MTP® (12f):	FC^12-3S
6 SC to MTP® (12f):	FC^12-3

SC Simplex Fiber Optic Connectors

MM (900µm/3mm, black):	FSCMBL
MM (900µm/3mm, red):	FSCMRD
SM (900µm/3mm, blue):	FSCSBU
MM (900µm/2mm, black):	FSCM2.0BL
MM (900µm/2mm, red):	FSCM2.0RD
SM (900µm/2mm, blue):	FSCS2.0BU

SC Duplex Fiber Optic Connectors

Multimode (3mm only):	FSCDMM
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Opti-Core® Fiber Optic Patch Cords

Duplex SC to SC:	F^D3-3M‡
Simplex SC to SC:	F^S3-3M‡
Simplex SC to pigtail:	F^B3-NM‡

HD Connector Removal Tool

For LC and SC:	HDCRT
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**Substitute for module color:

EI = Electric Ivory	BU = Blue
IG = International Gray	WH = White
TG = Technical Gray	BL = Black
AW = Arctic White	IW = Off White
EW = European White	

^Substitute for fiber type:

X (OM3/OM4 – 10Gig™ 50/125µm),
5 (OM2 – 50/125µm), 6 (OM1 – 62.5/125µm)
or 9 (OS1 – 9/125µm).

‡Substitute for length in meters: 1, 2, 3, 5 or 10 for patch cords, and 1, 2 or 3 for pigtails; singlemode patch cords are also available in 12, 15, 25 or 30 meter lengths. Contact Customer Service for other available lengths.

SC Fiber Optic Adapters

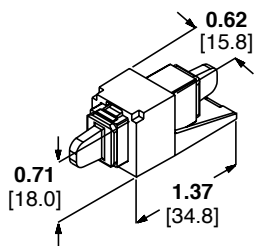
selection information

Part Number	Adapter Type	Split Sleeve Material	Adapter Color	Module Color	Average Insertion Loss^	Return Loss^
SC Adapters						
FASSCEI-L	Multimode Simplex	Phosphor Bronze	Electric Ivory	N/A	0.1dB	>20dB
FADSCEI-L	Multimode Duplex	Phosphor Bronze	Electric Ivory	N/A		
FASSCAQ-L	10Gig™ Multimode Simplex	Phosphor Bronze	Aqua	N/A		
FADSCAQ-L	10Gig™ Multimode Duplex	Phosphor Bronze	Aqua	N/A		
FASSCZAQ-L	10Gig™ Multimode Simplex	Zirconia Ceramic	Aqua	N/A		
FADSCZAQ-L	10Gig™ Multimode Duplex	Zirconia Ceramic	Aqua	N/A		
FASSCZBU-L	Singlemode Simplex	Zirconia Ceramic	Blue	N/A	0.15dB	>40dB
FADSCZBU-L	Singlemode Duplex	Zirconia Ceramic	Blue	N/A		
FASSCZAG-L	APC Singlemode Simplex	Zirconia Ceramic	Green	N/A		
FADSCZAG-L	APC Singlemode Duplex	Zirconia Ceramic	Green	N/A		
SC Adapter Modules						
CMSEISCEI	Multimode Simplex	Phosphor Bronze	Electric Ivory	Various**	0.1dB	>20dB
CMDEISCEI	Multimode Duplex	Phosphor Bronze	Electric Ivory	Various**		
CMSAQSCBL	10Gig™ Multimode Simplex	Phosphor Bronze	Aqua	Black		
CMDAQSCBL	10Gig™ Multimode Duplex	Phosphor Bronze	Aqua	Black		
CMSAQSCZBL	10Gig™ Multimode Simplex	Zirconia Ceramic	Aqua	Black		
CMDAQSCZBL	10Gig™ Multimode Duplex	Zirconia Ceramic	Aqua	Black		
CMSBUSCZBU	Singlemode Simplex	Zirconia Ceramic	Blue	Various**	0.15dB	>40dB
CMDBUSCZBU	Singlemode Duplex	Zirconia Ceramic	Blue	Various**		
CMSAGSCZBL	APC Singlemode Simplex	Zirconia Ceramic	Green	Black		
CMDAGSCZBL	APC Singlemode Duplex	Zirconia Ceramic	Green	Black		

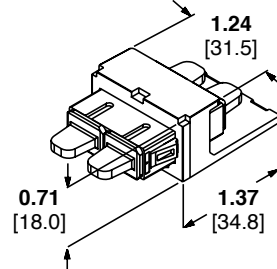
[^]Panduit SC Connector/Adapter connection

^{**}For other colors, replace suffix EI or BU with EI (Electric Ivory), BU (Blue), IG (International Gray), WH (White), TG (Technical Gray), BL (Black), AW (Arctic White), IW (Off White) or EW (European White).

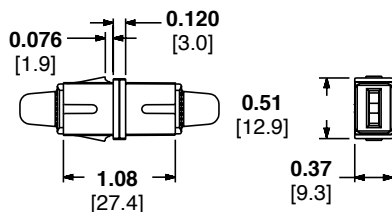
Mini-Com® SC Simplex Adapter Module



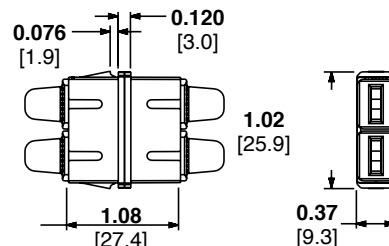
Mini-Com® SC Duplex Adapter Module



SC Simplex Adapter



SC Duplex Adapter



Dimensions are in inches [Dimensions in brackets are in millimeters]

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For more information

Visit us at www.panduit.com

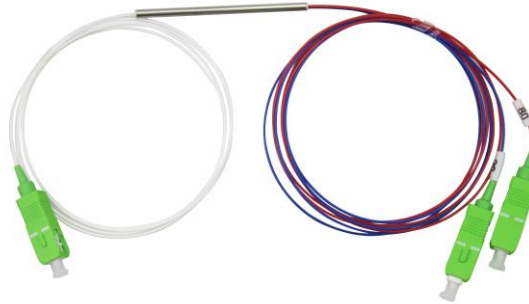
Contact Customer Service by email: cs@panduit.com
or by phone: 800-777-3300

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Single-mode FBT Coupler Datasheet

Widely used in optical network monitoring, optical fiber sensing, FTTX, CATV and other fields.



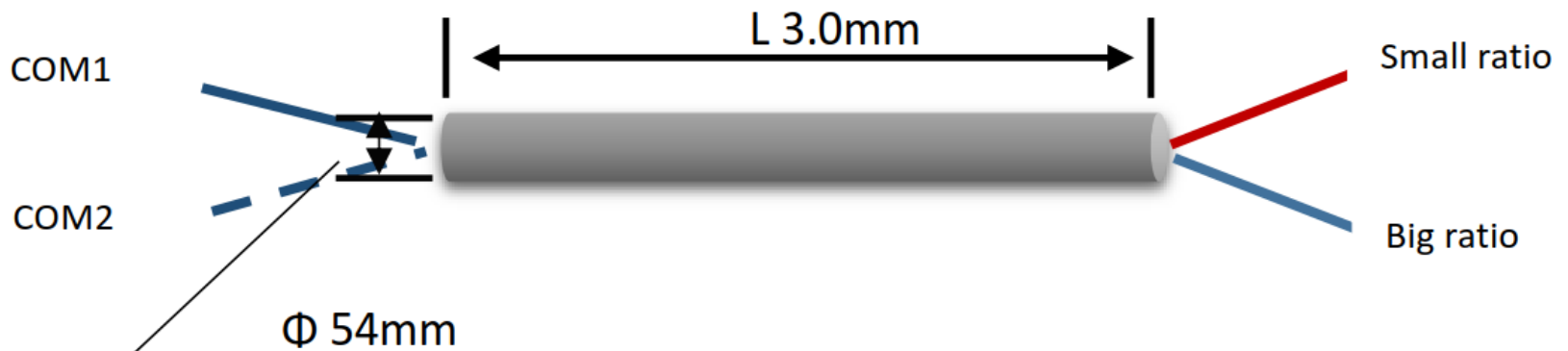
• Description

The single-mode FBT coupler is used to fuse two or more optical fibers under the condition of high-temperature heating and stretch them to both sides, PON network is mainly used for optical signal distribution and monitoring in optical networks and is one of the most important passive components. OPTICO offers a single-mode taper coupler with 1x2, 2x2 and a variety of spectral ratios to meet a variety of customer needs.

• Features

- Low insertion loss
- High stability and reliability
- Wide working wavelength range
- Compact design
- High stability and reliability
- Meets Telcordia GR-1209 and GR-1221, meets ROHS standard
- 100% tested before delivery
- Mature production technology

• Coupler drawing (Unit: mm):



• Optical Specifications:

Configuration		Unit	1x2 or 2x2			
Package Type		-	SSC	WBC	DWC	TWC
Operating Wavelength		nm	1310/1550	1310/1550	1310&1550	1310&1490&1550
Operating Bandwidth		nm	±20	±40	±40	1310&1550±40; 1490±10
Insertion loss	1/99	%	18.60~21.6/≤0.25	18.60~21.6/≤0.25	18.60~21.6/≤0.25	17.8~22.4≤0.35
	2/98		15.75~18.45/≤0.3	15.75~18.45/≤0.3	15.75~18.45/≤0.3	15.00~19.20/≤0.4
	3/97		14.1~16.5/≤0.35	14.1~16.5/≤0.35	14.1~16.5/≤0.35	13.6~17.05/≤0.45
	5/95		12.05~14.15/≤0.4	12.05~14.15/≤0.4	12.05~14.15/≤0.4	11.55~14.65/≤0.5
	10/90		9.20~11.2/≤0.75	9.20~11.2/≤0.75	9.20~11.2/≤0.75	8.8~11.4/≤0.8
	20/80		≤9.5/1.0	≤9.5/1.0	≤9.5/1.0	≤10.0/1.2
	25/75		≤7.9/1.3	≤7.9/1.3	≤7.9/1.3	≤8.4/1.4
	30/70		≤6.0/1.9	≤6.0/1.9	≤6.0/1.9	≤7.3/1.9
	33/67		≤5.7/2.2	≤5.7/2.2	≤5.7/2.2	≤6.3/2.1
	40/60		≤4.7/2.7	≤4.7/2.7	≤4.7/2.7	≤4.9/2.9
	45/55		≤4.2/3.2	≤4.2/3.2	≤4.2/3.2	≤4.4/3.4
	50/50		≤3.6	≤3.6	≤3.6	≤3.8
PDL		dB	≤0.15			
Return Loss		dB	≥50			
Directivity		dB	≥55			
Fiber type		-	SMF-28			
Temperature		°C	-40+85			
Package		Mm	3x45/3x54/3x60/90x20x10			
Connector Type		-	As customized			